

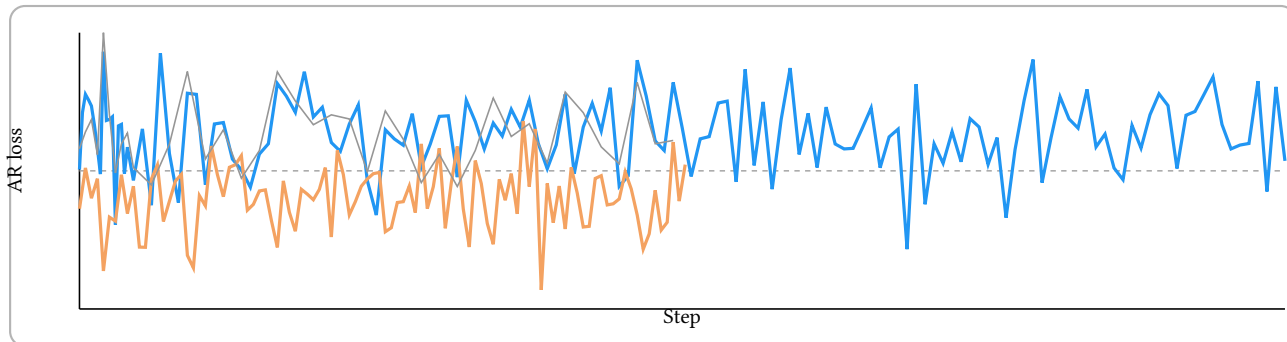
Results: Greedy Runs (135M vs 360M), draft length 8

Metric legend for 5 hour run, total cost 17.5 \$

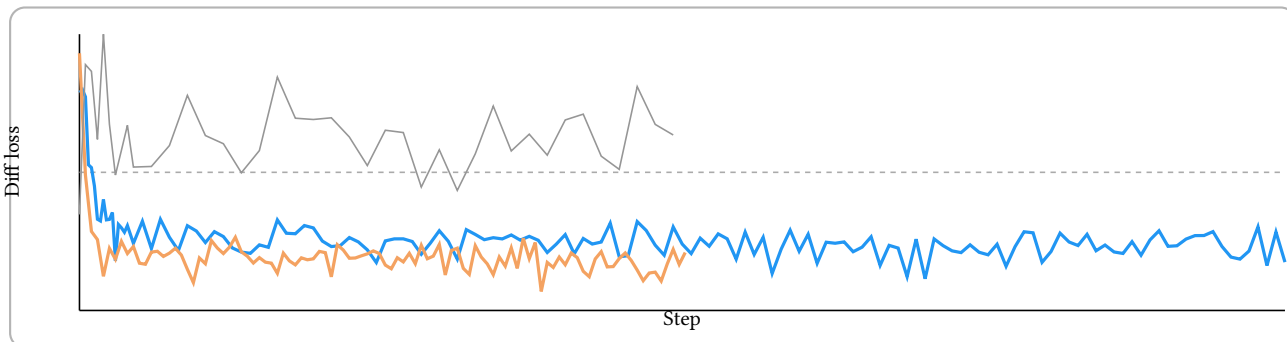
135M - 5.23\$ on 5090 with 32GB VRAM 360M - 11.81\$ on RTX PRO 6000 with 96GB VRAM

The bigger model ran with 50% bigger batch size but logs are synced by optimizer steps.

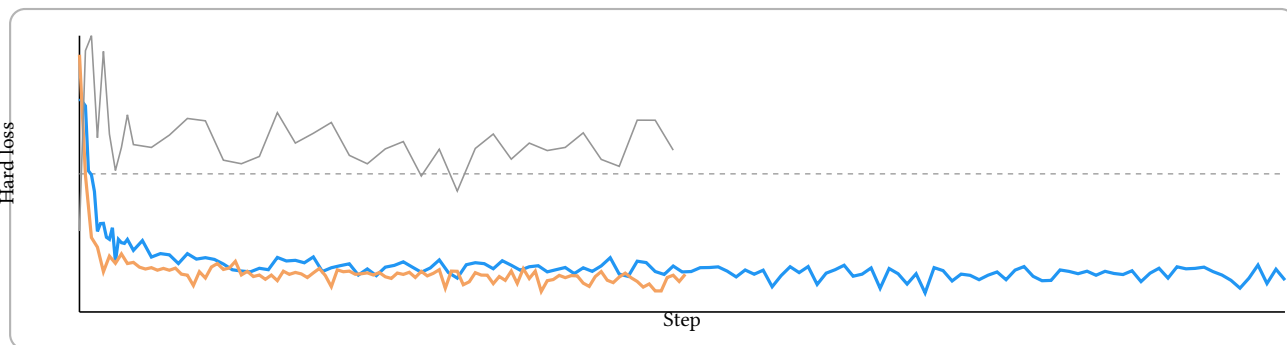
AR loss (lower is better) - This loss is not important for this experiment



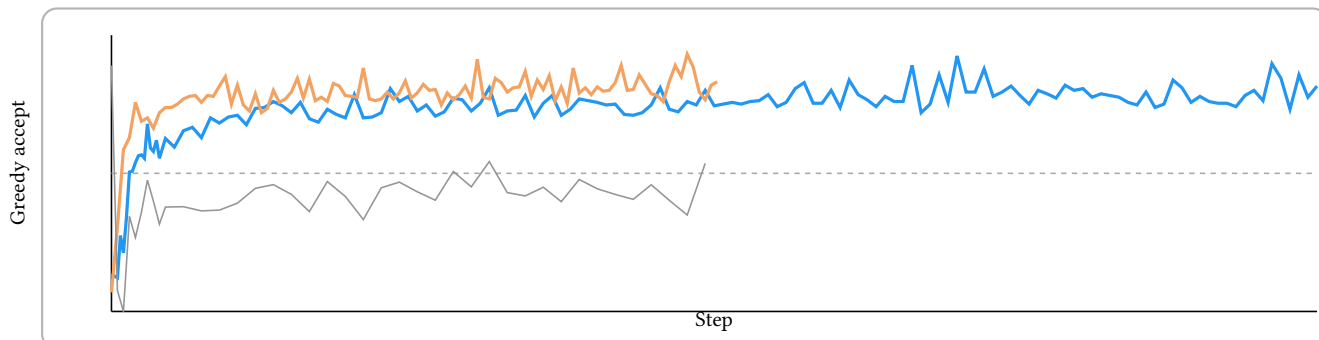
Diffusion loss (lower is better)



Hard (greedy) loss (lower is better)



Greedy acceptance (higher is better)



Inference (Once upon a time, verbose)

135M (step_0040000)

Output: “Once upon a time, John Grusky, the first White House astronomer, was chosen as a first-year intern. The first thing he saw was Mount Tai, in sight of the top of Mount Fresar in the Midwest. At first, Grusky didn’t think of it as a comet. He loved comparing it to the”

- **n_iterations: 44**
- **avg_accept_per_iter: 1.43**
- **max_accept_per_iter: 4**
- tokens_per_second: 5.50

360M (step_0020000)

Output: “Once upon a time, there lived a wicked queen who was such a horrible figure that she even had to have a cat to put off her spirit. There was a girl named Rapunzel, who loved to play in the forest. Her father, the King, had given up his kingdom for a beautiful princess who loved him dearly.”

- **n_iterations: 45**
- **avg_accept_per_iter: 1.40**
- **max_accept_per_iter: 4**
- tokens_per_second: 3.44

Verbose output from generation with the 135m model

prompt: Once upon a time
draft len: 8; max steps: 12

```
anchor: , -> there
currently verified draft: (,) | the | was | a | the | the | the | the
sample from verified draft: there | was | a | the | the | the | the | the
accept 1 of 8: there
select 1-th draft for next: there | was | a | a | the | the | the | the
```

```
anchor: there -> man
currently verified draft: ( there) | was | a | a | the | the | the | the
sample from verified draft: was | a | man | the | the | the | the | the
accept 3 of 8: was | a | man
select 3-th draft for next: man | named | a | the | . | the | the | the
```

```
anchor: man -> John
currently verified draft: ( man) | named | a | the | . | the | the | the
sample from verified draft: named | John | the | . | the | the | the | the
accept 2 of 8: named | John
select 2-th draft for next: John | . | . | He | was | a | a | .
```

```
anchor: John -> He
currently verified draft: ( John) | . | . | He | was | a | a | .
sample from verified draft: . | He | He | was | a | a | . | He
accept 2 of 8: . | He
select 2-th draft for next: He | was | was | a | a | , | , | ,
```

```
anchor: He -> a
currently verified draft: ( He) | was | was | a | a | , | , | ,
sample from verified draft: was | a | a | a | , | , | , | and
accept 2 of 8: was | a
select 2-th draft for next: a | man | man | man | man | He | He | He
```

```
anchor: a -> very
currently verified draft: ( a) | man | man | man | man | He | He | He
sample from verified draft: very | man | man | man | He | He | He | He
accept 1 of 8: very
select 1-th draft for next: very | man | He | He | He | He | He | He
```

final output

Once upon a time, there was a man named John. He was a very

Verbose output from generation with the 360m model

prompt: Once upon

draft len: 8; max steps: 12

anchor: a -> in

currently verified draft:	(a)		time		,		a		a		a		a		a
sample from verified draft:	time		,		in		a		a		a		a		a
accept 3 of 8:	time		,		in										
select 3-th draft for next:	in		was		a		a		a		a		a		the

anchor: in -> a

currently verified draft:	(in)		was		a		a		a		a		a		the
sample from verified draft:	a		a		a		a		a		a		the		a
accept 1 of 8:	a														
select 1-th draft for next:	a		small		,		,		,		,		,		,

anchor: a -> land

currently verified draft:	(a)		small		,		,		,		,		,		,
sample from verified draft:	land		,		,		,		,		,		,		town
accept 1 of 8:	land														
select 1-th draft for next:	land		town		,		,		,		,		a		a

anchor: land -> far

currently verified draft:	(land)		town		,		,		,		,		a		a
sample from verified draft:	far		,		,		,		,		a		a		man
accept 1 of 8:	far														
select 1-th draft for next:	far		the		,		,		,		,		a		a

anchor: far -> ,

currently verified draft:	(far)		the		,		,		,		,		a		a
sample from verified draft:	,		,		,		,		,		a		a		,
accept 1 of 8:	,														
select 1-th draft for next:	,		,		,		,		a		a		a		a

anchor: , -> far

currently verified draft:	(,)		,		,		a		a		a		a		a
sample from verified draft:	far		,		a		a		a		a		a		a
accept 1 of 8:	far														
select 1-th draft for next:	far		,		,		a		a		a		a		a

anchor: far -> away

currently verified draft:	(far)		,		,		a		a		a		a		a
sample from verified draft:	away		,		a		a		a		a		a		a
accept 1 of 8:	away														
select 1-th draft for next:	away		,		a		a		a		a		a		a

anchor: away -> there

currently verified draft:	(away)		,		a		a		a		a		a		a
sample from verified draft:	,		there		a		a		a		a		a		a
accept 2 of 8:	,		there												
select 2-th draft for next:	there		a		a		a		a		a		.		.

final output

Once upon a time, in a land far, far away, there